

3(a)

B1

The tesla is the uniform magnetic flux density which when acting at right angle to a straight conductor carrying a current of 1 ampere produces a force per unit length of 1 Newton per metre on the conductor.

Or

If a conductor carrying a current of 1 ampere is placed at right angles to a uniform magnetic field of flux density 1 Tesla, then the force per unit length on the conductor is 1 newton per metre.

3(b)(i)

B1

Charged particles moving perpendicular to a magnetic field will experience a resultant magnetic force perpendicular to its motion. Hence no work is done. By Newton's 2nd Law, the acceleration of the particles is in the same direction as the resultant force. The direction of the particles changes but not its speed. By Newton's 1st law, upon exit, the particles will move in a straight line with a speed of 4500 ms⁻¹.

B1

3(b)(ii)

B1

Magnetic force provides centripetal force for particle's circular motion

$$F = Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{Bq}$$

$$= \frac{(2.66 \times 10^{-26})(4500)}{(2 \times 10^{-3})(1.6 \times 10^{-19})}$$

$$= 0.374 \text{ m}$$

A1

3(b)(iii)

M1

$$v = r\omega$$

$$\omega = \frac{v}{r} = \frac{4500}{0.374}$$

$$= 12030 \text{ rad s}^{-1}$$

A1

